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Introduction

Introduction, History, Applications



Basics of Python

Input, Output, Variables, Data Types



Control Statements

Conditional Statements, Loops



Function

Definition, Return Statements, Arguments



Module

Import , Export



OOP

Class, Object, Inheritance



About Python

01

Python is general purpose, high level, interpreted, object oriented programming language

02

Python was created by **Guido van Rossum** in the late 1980s and early 1990s at Centrum Wiskunde & Informatica (CWI) in the Netherlands. Python 1.0, was released in **February 1991**.

03

Today, Python is maintained by the **Python Software Foundation (PSF)** with current version 3.13.1



Guido Van Rossum
(Microsoft)



Applications



Web Development

Django, Flask, Streamlit, FastAPI



Data Analytics And Visualization

Numpy, Pandas, SciPy, Matplotlib, Seaborn



Machine Learning & Artificial Intelligence

Scikit-learn, Tensorflow, Keras, Pytorch



Game Development

Pygame, Panda3D



Desktop GUI Applications

Tkinter, PyQt, wxPython, Kivy



Automation and Scrapping

Beautiful Soup, Scrapy, Selenium, requests



Installing Python

Step 1 : Download Python

Go to the official Python website: <https://www.python.org/downloads/>.

Step 2 : Run the Installer

Once the Python installer is downloaded, locate the installer in your "Downloads" folder and run it.

Important: On the installation screen, **check the box that says** "Add Python to PATH".

Click "Install Now".

Step 3 : Verify the Installation

After the installation completes, open Command Prompt (press Win + R, type cmd, and hit Enter).

Type the following command to check the Python version:

```
python version
```



Input And Output

The **input()** function in Python is used to receive input from the user during program execution. It pauses the program until the user provides input and presses Enter. The input is always returned as a string.

The **print ()** function is used to display the output to the console.

Different Ways Of Printing:

- Comma Separated method
- Using f-strings
- Using format method



variables

A variable in Python is a named reference to a memory location where data is stored. Variables make it easy to manipulate and access data throughout a program.

Variables are created when you assign a value to them. Python doesn't require explicit variable declarations or type specifications.

